

Aerodynamics III - 2025/26

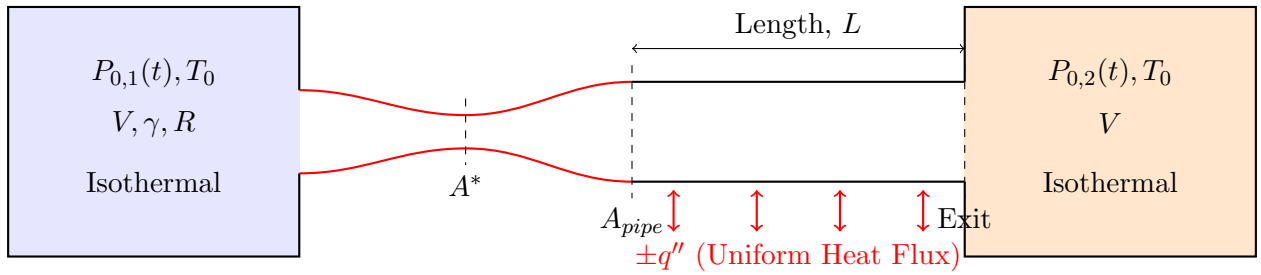
HW#1

Converging-Diverging Nozzle followed by a Pipe with Heat Transfer

Problem Description

Develop a computer program in Python language to compute the quasi-1D compressible flow for the analysis of a convergent-divergent nozzle connected to a constant cross-section pipe subjected to an imposed uniform heat flux.

Consider the fluid as inviscid and calorically perfect, with an imposed specific heat ratio (γ) and specific gas constant (R) defined by the user. The flow is isentropic in the converging-diverging nozzle section (with the exception of possible normal shocks). In the straight pipe, the flow is frictionless but non-adiabatic (Rayleigh flow conditions, where heat can be added or removed).



Tasks and Requirements

The project is divided into two main tasks: Task 1 and Task 2. For each task, you must develop an algorithm, coding it into a standalone application or script.

Task 1: Steady-State Analysis (Infinite Reservoirs) – 75% of Grade

Assume the reservoirs are infinitely large ($V \rightarrow \infty$), meaning stagnation pressures $P_{0,1}$ and $P_{0,2}$ are constant.

1. Develop an algorithm to solve the spatial fields ($y = f(x)$) for the entire domain.
2. The code must handle user inputs for $\gamma, R, P_{0,1}, P_{0,2}$, the shared stagnation temperature T_0 , nozzle geometry (as explicit functions of A^* and A_{pipe}), pipe length L , and the imposed uniform heat flux q'' (positive for heating, negative for cooling).
3. Provide graphical representation of the pertinent variables along the flow direction, e.g. $\dot{m}, T(x), T_{0,x}(x), P(x), P_{0,x}(x), M(x), s(x)$.
4. **Implement physical safeguards for predictable flow adjustments.** Specifically, the code should predict and handle the occurrence of Normal Shock Waves (NSW) arising from the imposed back-pressure ($P_{0,2}$) or from thermal choking (due to high q'').

Important Note: If your group is unable to fully code the exact prediction and resolution of the NSW, you may opt to simply identify when/where an NSW would occur and stop the code. **However, taking this simplified route will result in a grading penalty for Task 1, and you will not be able to proceed to Task 2.**

Task 2: Transient Analysis (Finite, Isothermal Reservoirs) – 25% of Grade

Consider the two reservoirs to have finite, equal volumes (V) and the same constant stagnation temperature (T_0). Their stagnation pressures will change over time as mass transfers from Reservoir 1 to Reservoir 2.

1. Apply the mass balance to estimate the time required from the initial state until the system reaches the final equilibrium state ($P_{0,1} = P_{0,2}$).
2. Assume that the pipe system (analyzed in Task 1) operates under **quasi-steady state**, i.e. negligible fluid inertia, hence, the flow adapts instantly to the current reservoir pressures. Note that this assumption is realistic only if the volume of the reservoirs (V) is much larger than the total volume of the converging-diverging nozzle plus the pipe. Otherwise, although the model can be applied, it will generate inaccurate predictions.
3. Discretize the mass balances using a simple explicit numerical method in time.
4. **Outcomes:** The user interface must output the time-evolution of the stagnation pressures in the reservoirs ($P_{0,1}(t)$ and $P_{0,2}(t)$) and the mass flow rate ($\dot{m}(t)$), as well as the evolution of key variables (e.g. M , P , T) over time at specific points (throat, nozzle inlet/outlet, and pipe inlet/outlet).

Evaluation Criteria

The HW's evaluation considers:

- A concise report with a **maximum of 5 pages**, using the strict report template provided (please keep the figures/pictures readable). *Note: Many specific reporting details, cases, and required plots are clarified within the template itself.*
- Ease of graphical input and output, and correctness of results.
- The allowable range of inputs and robustness of programmed safeguards (handling of thermal choking and shock waves).
- A critical physics analysis of the fields for specific test conditions (Task 1) and a discussion of the transient blowdown process (Task 2).
- **Submission:** The submission must be done through Google Forms (link below) until the date **16/03/2026 (END OF THE DAY)**. Students must submit **exactly two files**:
 1. The Report PDF (max 1 MB) with file name 'HW1_report_group_##.pdf' (the group number ## should have 2 digits, e.g. for group 1 consider ## = 01).
 2. A ZIP file containing the report's LaTeX source and the Python codes ready to run (max 1 MB); file name 'HW1_report_group_##.zip' (the group number ## should have 2 digits, e.g. for group 1 consider ## = 01). *To save space, do NOT provide output files or compiled executables.*
 3. google forms link: <https://forms.gle/QPP5BHnp3CkEXiZA9>

- **Discussion:** After the report submission, a date will be defined for a 15-minute discussion based on the report results. During this session, students must run the codes for Tasks 1 and 2, reproduce the report results, clarify details about the developed code, and discuss the physical outcomes.

Remarks

- **Code base:** The code and application **must be developed in Python**. You are encouraged to use libraries like NumPy or SciPy for numerical calculations, and libraries such as Tkinter, PyQt, PySide, or Streamlit for the Graphical User Interface (GUI).
- **Originality:** Your programming contribution should be highlighted. All parts of the code obtained from libraries, toolboxes, or other parties must be referred to as external code. Similarity checks will be conducted using URKUND. Plagiarism, if detected, may be reported!
- **Flowchart:** The report should contain flowcharts with the algorithm (as requested in the report template).
- **Format:** Produce an easy-to-run Python application with a GUI. Due to the strict 1 MB file size limit for submission, **do not compile the script into a standalone executable** (e.g., avoid PyInstaller). Ensure your code can be executed easily from the source files using standard libraries. Bonus points can be obtained for out-of-the-box exploration.
- **Submission:** Strictly follow the 2-file rule detailed in the Evaluation Criteria section.